



ACID BASE IMBALANCE & ABGs

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Acid/Base Balance



Normal Acid/Base balance of blood pH is 7.35 to 7.45



Proper pH balance is necessary for cellular function, optimal enzymatic activity, and to maintain homeostasis



Regulated by the Renal and Respiratory Systems (Metabolic vs Respiratory)

Respiratory Acidosis

Pathophysiology:

- Excessive retained CO₂ (>45) & pH is Acidic (<7.35)
Usually a result of hypoventilation

Etiology:

- Obstructive pulmonary diseases (COPD, emphysema, asthma)
Respiratory depression (head injury, CVA, drug OC)

Signs & Symptoms

- SOB, dyspnea, headache, tachycardia, HTN, hypoxia, cyanosis, drowsiness, confusion, obtunded

Interventions:

- Monitor respiratory status, O₂ Sats

Correct underlying condition

- O₂ therapy, mechanical ventilation if necessary, Narcan if opioid overdose, ABX for PNA

Respiratory Alkalosis

Pathophysiology:

- CO₂ is exhaled in excess (<35) & pH is alkaline/base >7.45. Usually a result of hyperventilation

Etiology:

- Anxiety, panic attacks, stress, hypoxia

Signs & Symptoms

- Dizziness, lightheadedness, paresthesia, palpitations, SOB, chest tightness, difficulty concentrating, confusion

Interventions:

- Breathe into paper bag to re-breathe CO₂

Correct underlying condition

- Monitor for signs of hyperventilation, low O₂, hypocalcemia (muscle cramps, tetany)



Metabolic Acidosis

Pathophysiology:

- Overproduction or inability for kidneys to filter out
- acid &/or loss of HCO_3 (<22) & pH is acidic <7.35 .

Etiology:

- Overproduction (DKA, lactic acidosis); Low HCO_3 (diarrhea, renal failure, CKD); Impaired renal funx (CKD or AKI/ARF)

Signs & Symptoms

- Headache, weakness, fatigue, NVD, low BP, tachycardia, confusion, disorientation

Interventions:

- Monitoring RR, depth, O_2 & vital signs, mental status, fluid balance

Correct underlying condition

- Restore electrolyte balance, HCO_3 if pH is critically low, HD if needed, Insulin for DKA

Metabolic Alkalosis



Pathophysiology:

- Increase in Bicarb ($\text{HCO}_3^- > 26$) or loss of Hydrogen ions & pH is > 7.45 (Base/Alkaline)

Etiology:

- Loss of acids (vomiting, NG suction); Increase in HCO_3^- (antacids or diuretics); Endo disorders (Hyperaldosteronism, Cushing's syndrome)

Signs & Symptoms

- Nausea, Vomiting, Muscle twitch/cramps, tetany, low BP, confusion, agitation, respiratory depression (from compensation), confusion/agitation

Interventions:

- Monitor labs, respiratory rate, promote hydration, correct electrolyte imbalances (K or Cl replacement)

Correct underlying condition

- Correct hyperventilation, low O_2 , hypocalcemia (muscle cramps, tetany), D/C diuretics &/or other contributing meds

Common Electrolyte Imbalances for each Acid/Base Imbalance

Metabolic Acidosis:

- **Hyperkalemia:** As the blood pH drops (acidic), potassium shifts from the intracellular space to the extracellular space (outside the cells), leading to higher potassium levels in the blood.
- **Hyponatremia:** Often seen in the setting of diabetic ketoacidosis (DKA) or renal failure, where water retention or dilutional effects can lower sodium levels.
- **Hypocalcemia:** Acidosis can cause a decrease in protein binding of calcium, leading to an increase in ionized calcium, although total calcium may be low.
- **Hyperchloremia:** Chloride may rise in conditions of normal anion gap metabolic acidosis, such as diarrhea, or remain normal in high anion gap acidosis, like diabetic ketoacidosis or renal failure.

Respiratory Acidosis:

- **Hyperkalemia:** Like metabolic acidosis, respiratory acidosis leads to a shift of potassium from the intracellular space to the extracellular space due to the acidic environment.
- **Hypocalcemia:** A rise in carbon dioxide levels can decrease ionized calcium by increasing protein binding.
- **Hyperchloremia:** May increase in cases of chronic respiratory acidosis due to renal compensation for bicarbonate retention.

Metabolic Alkalosis:

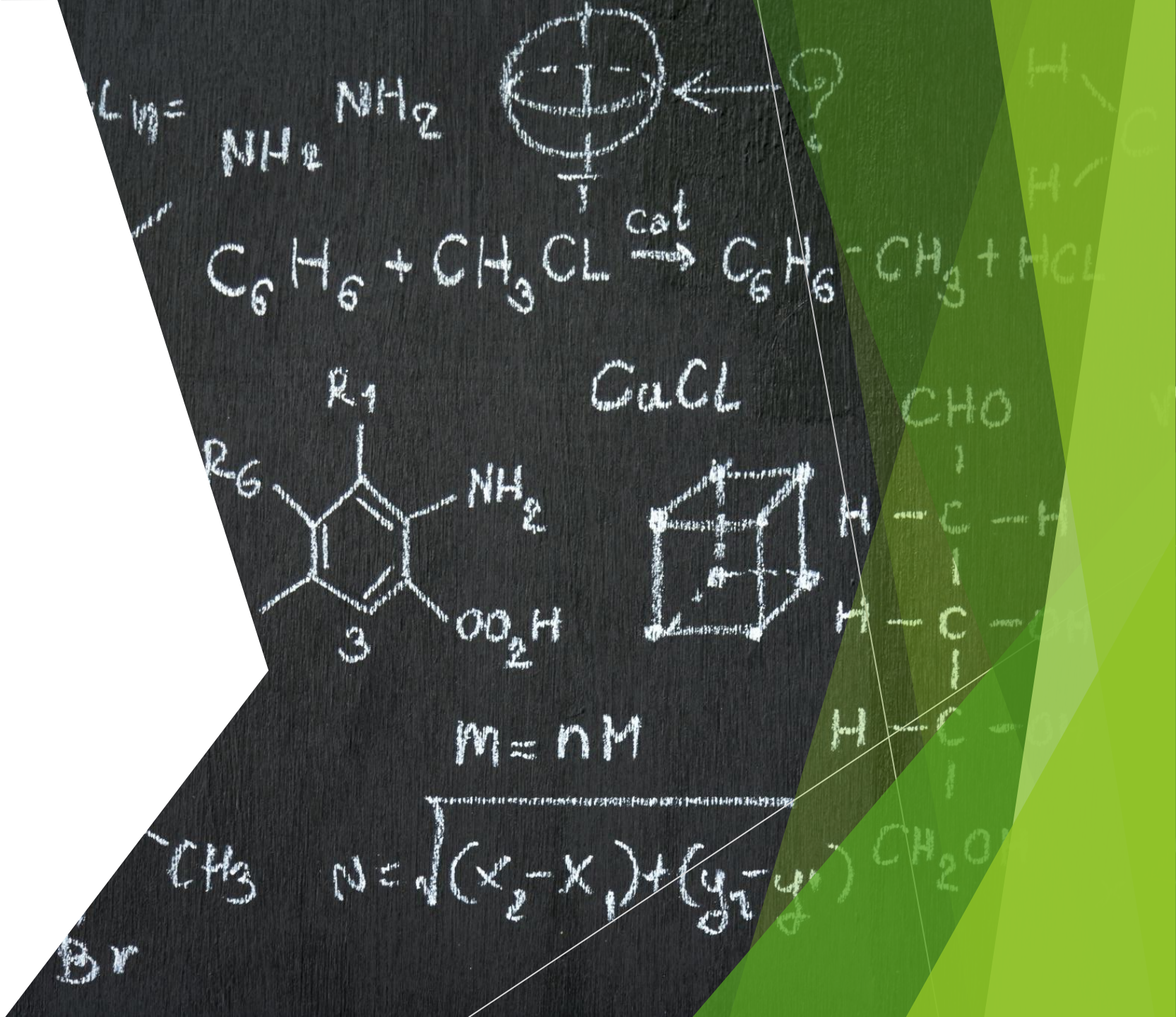
- **Hypokalemia:** As the blood pH increases (alkalotic), potassium shifts into cells and renal losses of potassium often occur.
- **Hypernatremia:** if alkalosis is caused by GI losses leading to dehydration, the increased concentration of sodium leads to hypernatremia in the blood stream.
- **Hypocalcemia:** Alkalosis can cause an increase in protein binding of calcium, leading to a decrease in ionized calcium, although total calcium often remains normal.
- **Hypochloremia:** elevated bicarb increases renal excretion of chloride to help maintain electrolyte balance.

Respiratory Alkalosis:

- **Hypokalemia:** Respiratory alkalosis often leads to potassium shifting into cells, causing a decrease in blood potassium levels.
- **Hypocalcemia:** As with metabolic alkalosis, the increased pH leads to more calcium binding to proteins, which can lower ionized calcium levels.
- **Hypophosphatemia:** Respiratory alkalosis can increase phosphate excretion, potentially lowering phosphate levels in the blood.

So... what are we doing with all these numbers????

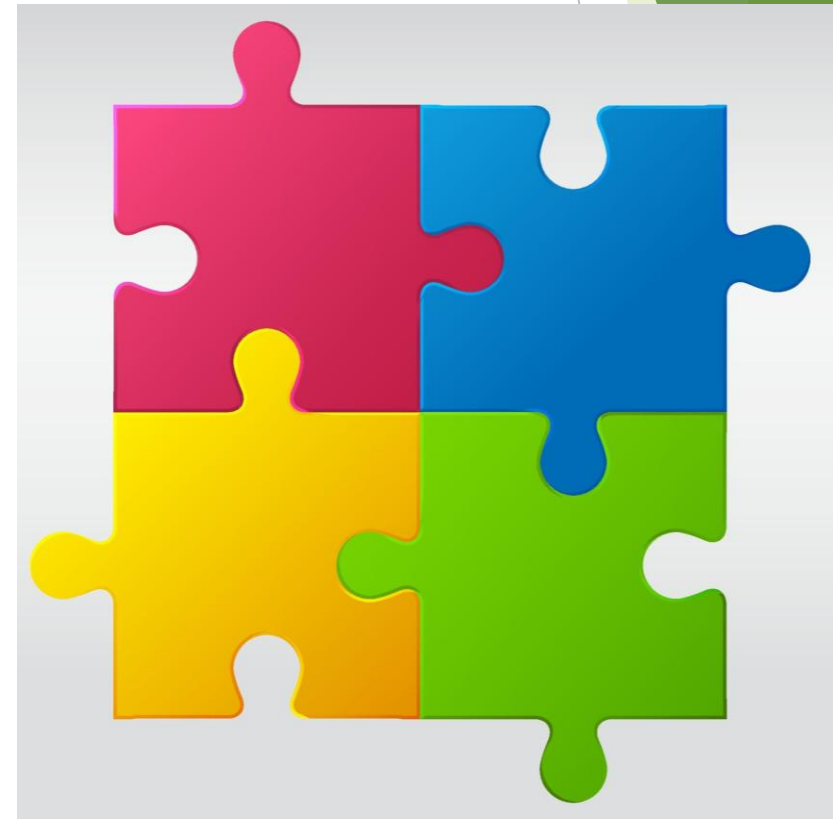
- What do the numbers mean?
- Which of the 4 imbalances are we seeing?
- What causes THAT imbalance?
- Medical/Nursing Interventions?
- Monitoring?



ABGs are NOT HARD!!

- ▶ They are not hard to calculate
- ▶ They are not hard to figure out
- ▶ Compensation is not hard

- ▶ They are “tricky” but if you think of them like a puzzle, they can be FUN :D
- ▶ If you know that MOST nurses don’t know how to interpret them, it’s also bragging rights? (j/k we’re always humble ♥)



ABGs - Arterial Blood Gasses

- ▶ An arterial blood gas (ABG) test measures the oxygen and carbon dioxide levels in your blood as well your blood's pH balance.
- ▶ Often ordered when patient is in distress
- ▶ Blood sampled from an **ARTERY** – often from the Radial artery on the wrist
- ▶ Can be drawn by RN or RT – at **ONVIDA** it is **RT** (and RNs are not trained for arterial sticks)
- ▶ Must hold pressure for much longer than venipuncture to achieve hemostasis & higher risk for bleeding after the procedure. NO BPs on that arm for 24 hrs!!



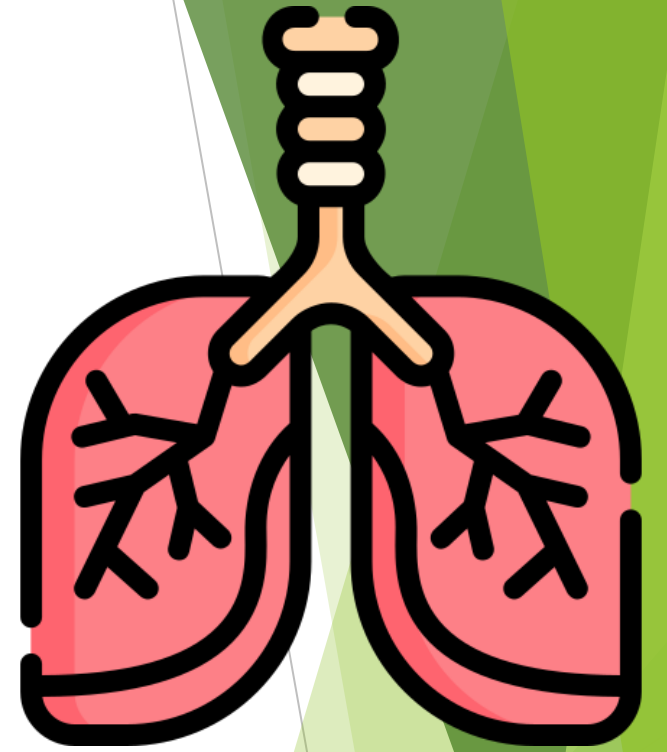
pH = “Power of Hydrogen”

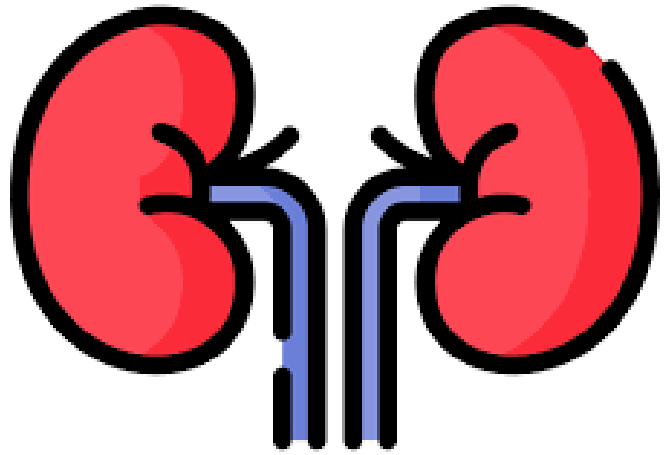


Carbon Dioxide (CO_2)

think acid! & only the lungs
can “blow it off”

MORE CO_2 = MORE ACID





Bicarb (HCO_3)

think of it as a BUFFER for acid & the kidneys regulate it

<-----22-----26----->

ACID

NORMAL

BASE

LESS BICARB TO BUFFER THE ACID

Interpreting ABGs - KISS

(Keep It Simple Smarties)

- ▶ **1.) Is pH acid normal or base?**
 - ▶ Is CO₂ acid normal or base?
 - ▶ Is HCO₃ acid normal or base?

- ▶ **2.) Determine which of the 2 are contributing to the pH abnormality.**
 - ▶ If pH is acidic, whichever is acidic is causing it.
 - ▶ If pH is alkaline, whichever is alkaline is causing it.

- ▶ **3.) Once you decide ACIDIC or ALKALINE, determine if the other body system is trying to help:**
 - ▶ If pH is acidic & CO₂ is acidic but the HCO₃ is alkaline, THE KIDNEYS ARE HELPING
 - ▶ If pH is alkaline & CO₂ is alkaline but the HCO₃ is acidic, THE KIDNEYS ARE HELPING
 - ▶ If pH is acidic & HCO₃ is acidic but the CO₂ is alkaline, THE LUNGS ARE HELPING
 - ▶ If pH is alkaline & HCO₃ is alkaline but the CO₂ is acidic, THE LUNGS ARE HELPING

ABG Interpretation

pH



CO₂

RESPIRATORY



HCO₃

METABOLIC
(KIDNEYS)



Figuring it out...

► When no one is helping... ☹️

- If pH is acidic & CO₂ is acidic but the HCO₃ is normal
- RESPIRATORY ACIDOSIS
- If pH is alkaline & CO₂ is alkaline but the HCO₃ is normal
- RESPIRATORY ALKALOSIS
- If pH is acidic & HCO₃ is acidic but the CO₂ is normal
- METABOLIC ACIDOSIS
- If pH is alkaline & HCO₃ is alkaline but the CO₂ is normal
- METABOLIC ALKALOSIS

If EVERYTHING is acid, it is both Respiratory and Metabolic Acidosis

If EVERYTHING is base, it is both Respiratory and Metabolic Alkalosis

Compensation... How the body works to BALANCE

- ▶ The body will try to balance things out to get the pH back to normal.
- ▶ The other body system should begin to help (aka “compensate”)
- ▶ Changes in Bicarb (HCO_3) happen slower in the renal system.
 - Increases and decreases in HCO_3 excretion take longer but have a longer effect.
- ▶ Changes in CO_2 from the respiratory system happen much faster.
 - Increases and decreases in CO_2 can affect change faster
 - Increased RR & depth will blow off more CO_2 (“less acid”)
 - Decreased RR &/or depth will blow off less CO_2 (“more acid”)
 - We can voluntarily BUT temporarily affect both mechanisms.
- ▶ This help is either PARTIAL w/an abnormal pH or FULL w/a normal pH.



When the body is trying to balance things out: **compensation...**

- ▶ If **pH is acidic** & **CO₂ is acidic** but the **HCO₃** is alkaline, **THE KIDNEYS ARE TRYING TO HELP**
- ▶ **RESPIRATORY ACIDOSIS** WITH **METABOLIC COMPENSATION**

- ▶ If **pH is alkaline** & **CO₂ is alkaline** but the **HCO₃** is acidic, **THE KIDNEYS ARE TRYING TO HELP**
- ▶ **RESPIRATORY ALKALOSIS** WITH **METABOLIC COMPENSATION**

- ▶ If **pH is acidic** & **HCO₃ is acidic** but the **CO₂** is alkaline, **THE LUNGS ARE TRYING TO HELP**
- ▶ **METABOLIC ACIDOSIS** WITH **RESPIRATORY COMPENSATION**

- ▶ If **pH is alkaline** & **HCO₃ is alkaline** but the **CO₂** is acidic, **THE LUNGS ARE TRYING TO HELP**
- ▶ **METABOLIC ALKALOSIS** WITH **RESPIRATORY COMPENSATION**

FULL compensation...

► When there is help.. 😊

If pH is normal but the CO_2 and/or HCO_3 are abnormal,
then the original acid/base imbalance is *FULLY COMPENSATED*

Remember this phrase:

“Fully Compensated To A Normal pH”

The compensation was enough help

BALANCE the pH

BACK TO NORMAL

What do you need to know?

As a nurse... For an exam...

How to calculate the ABGs & interpret the abnormalities

Then what are you going to do about it?

▶ Metabolic Acidosis

- ▶ Causes, S&S, Interventions to correct it

▶ Respiratory Acidosis

- ▶ Causes, S&S, Interventions to correct it

▶ Metabolic Alkalosis

- ▶ Causes, S&S, Interventions to correct it

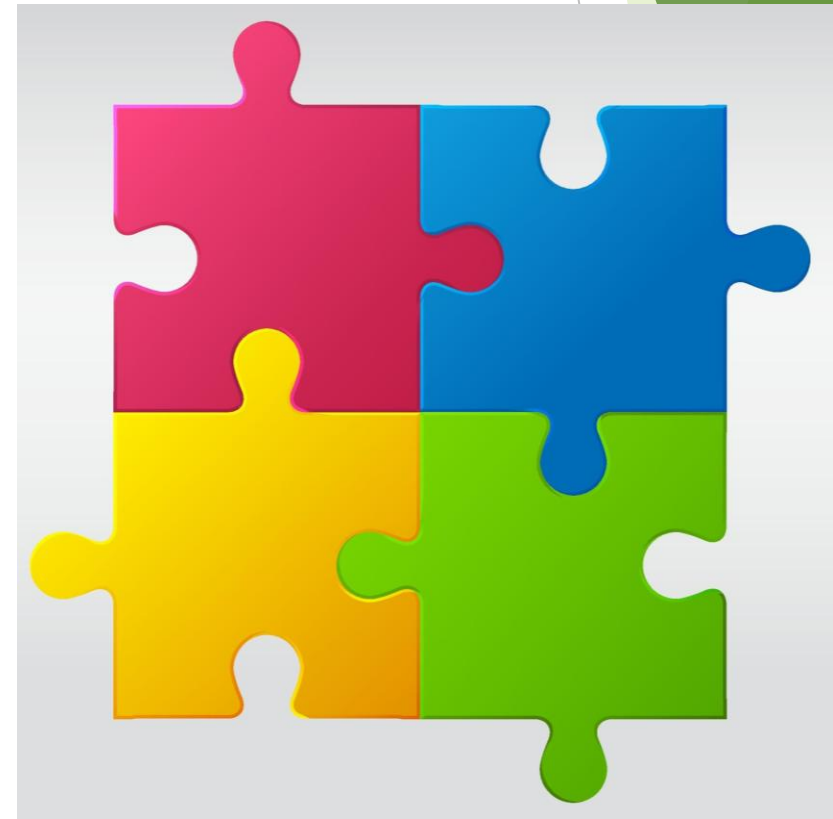
▶ Respiratory Alkalosis

- ▶ Causes, S&S, Interventions to correct it

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THANK YOU!